

Class 19: The Confidence Game -- Instructor Notes

Bootstrap CI for a mean, and for a regression slope

Overview

50 minutes. Files in `Class19/`, `data = weight-height.csv` (Part 1, your real file – 10,000 rows, Gender/Height/Weight, height in **inches**) and `cricket_thermometer.csv` (Part 2, same 15-row Pierce dataset as Class 18 – verified to reproduce Class 18's numbers exactly).

Note: `weight-height.csv` is the classic Kaggle Gender/Height/Weight set: 5,000 Male + 5,000 Female rows, height in inches (not cm – the original warm-up notebook's print label said "cm"; corrected here). `population.select("Height")` pools both genders, so the population is mildly bimodal (Male mean 69.03 in, SD 2.86; Female mean 63.71 in, SD 2.70). This doesn't break anything – the bootstrap CI for a mean doesn't require a normal population – and the $n=10/n=500$ histograms below come out visibly unimodal regardless. Worth having the fact in your pocket in case a sharp student asks why the population itself isn't bell-shaped.

Arc: replicate-vs-resample on a die roll (why bootstrap works at all) – worked height-mean example – predict-first before scaling $n=10$ to $n=500$ – callback to Class 18's cricket slope – predict-first before bootstrapping that slope – spaghetti-plot visualization – adapt to the intercept, which turns out to be a second, unplanned lesson about extrapolation – discussion of when the method breaks.

Timing

Part	Min	Content
1	15	Die-roll bootstrap (concept) -> height mean, $n=10$ -> predict -> $n=500$
2	20	Cricket slope bootstrap (worked) -> spaghetti plot -> adapt to intercept
3	3	Bridge to Lab 09
4	10	Paper discussion – assumptions and limits
Total	48	2 min buffer

Part 1 – Key numbers (verified against your real data)

Note: Die-roll bootstrap validates the method itself: the replicate-many-times CI and the resample-from-one-sample CI should land close to each other. In one test run: replication CI [3.020, 3.980], bootstrap CI [2.740, 3.760] – close, as expected (both runs are random, so exact bounds will shift each time; the point is the two methods agreeing, not the specific numbers). This is worth saying out loud to students: **we're not doing anything magical, we're just checking that resampling-from-one-sample gives approximately the same answer as actually replicating the experiment** – which is the entire justification for trusting the bootstrap later, when replication isn't possible.

Population mean height (both genders pooled): 66.37 in. One test draw:

n	95% CI	Width
10	63.93 – 67.69	3.77
500	65.99 – 66.65	0.66

Ratio of widths 5.7x for a 50x increase in n. Theory says $\sqrt{50} \approx 7.07$; a single random draw of the n=10 sample won't hit that exactly (this is itself worth naming if a student computes a different ratio than a neighboring team – everyone drew a different random n=10 sample). Point for discussion (Q1.5): CI width scales with $\frac{1}{\sqrt{n}}$, not $\frac{1}{n}$ – quadrupling precision costs 16x the data, not 4x.

Note: Because this cell re-randomizes every run, exact bounds will differ slightly each time you or a student executes the notebook – including between your instructor run and what students see live. That's expected and itself worth a sentence: the **CI** is a range estimate, not a single fixed number, so a small amount of run-to-run jitter in **where exactly** it lands is the method working correctly, not a bug.

Part 2 – Key numbers (verified against the real Class 18 dataset)

Note: The cricket data reproduces Class 18's stats exactly: mean temp 80.04°F, slope 3.2911, intercept 25.2323, $r = 0.8351$, $r^2 = 0.6975$. This is the same `cricket_thermometer.csv` – no new data file needed if it's already sitting in your repo from Class 18.

Quantity	Point estimate	95% bootstrap CI
Slope	3.2911	2.27 – 4.54
Intercept	25.2323	3.7 – 43.5

Caution: The intercept's CI is **enormous** relative to its point estimate – almost an order of magnitude wider, proportionally, than the slope's. This is not a bug. The intercept is the predicted temperature at 0 chirps/sec, far outside the observed range of 14.4–20 chirps/sec, so resampling amplifies its uncertainty the same way Class 18's negative- R^2 extrapolation did. If a team gets a wildly wide intercept CI and asks whether they made a mistake, the answer is no – that width **is** the finding. Worth flagging explicitly in Q2.4 debrief, and a clean callback to “predictions get shakier the further you extrapolate” from last week.

Does the slope CI include 0? No – comfortably clears it (2.27 is well above 0), consistent with $r = 0.835$. This is worth connecting explicitly to hypothesis testing: a CI that excludes 0 is equivalent to rejecting the null of “no relationship” at the 5% level. Some students will make this connection unprompted; if not, it's a good thing to surface in the Q2.2 debrief.

Part 4 – Discussion guidance

Q4.1 (assumptions). Target answer: the sample needs to be reasonably representative / randomly drawn from the population of interest. A bad-case example for crickets: all 15 measured from the same colony on the same night (temperature range too narrow, pseudo-replication). For height: sampling only one gender, or one age group, would make the “sample stands in for the population” assumption fail even though the code runs fine and produces a confident-looking interval.

Q4.2 (200 vs. 10,000 reps). This is the Lab 7 lag-2 callback from Class 14, restated: a narrower interval isn't automatically a better one if it came from fewer resamples – with only 200 reps, percentile estimates are noisier, and a “tighter” result is as likely to be bad luck as it is to be a better estimate. More reps reduces **simulation** noise around a fixed answer; it does not fix **sampling** noise from having only 15 crickets to begin with. Those are two different sources of uncertainty and it's easy for students to conflate them – worth drawing that distinction explicitly if it doesn't come up.

Q4.3 (n=15 vs n=500 resample composition). With n=15, a single resample easily omits an influential point (e.g., the 93.3°F / 19.8 chirps observation) or triples it up, visibly shifting the fitted line – that's exactly what the spaghetti plot in Part 2 shows. With n=500, no single point has enough leverage to move the line much regardless

of how many times it's resampled. This is the mechanical reason small-n bootstrap CIs are wide and lumpy while large-n ones are narrow and smooth.

Open items

- **Font fallback** – same Liberation Serif issue noted for Classes 16–18. Unresolved; left unchanged.
- **preamble.typ is a reconstruction**, not your real preamble – paste yours over it before compiling for real.
- **weight-height.csv is now your real file** – verified above (10,000 rows, inches, mixed gender). No further action needed.
- **Cricket_Thermometer.ipynb item closed** – its content (bootstrap CI for slope, seaborn-style visualization) is now folded into this class via the spaghetti plot in Part 2; no separate notebook needed.